- *Disaster management and assessment*: Drones equipped with optical and thermal sensors were deployed for search and rescue, damage assessment, and resource mapping following natural disasters such as earthquakes, floods, and hurricanes.
- *Environmental monitoring*: UAVs contributed to wildlife tracking, forest health assessments, and deforestation mapping, supporting ecological research and conservation projects.
- *Oceanographic and polar research*: Endurance-capable UAVs collected data on sea ice movement, ocean currents, and polar atmospheric patterns, aiding climate change modeling.

4.4.3 Civilian and commercial applications

Third-generation drones extended far beyond defense, finding widespread adoption in civilian, industrial, and infrastructure-related applications due to their precision, efficiency, and cost-effectiveness.

- *Aerial mapping and surveying*: Equipped with high-resolution cameras and LiDAR systems, UAVs were used for urban planning, topographical mapping, and construction monitoring with centimeter-level accuracy.
- *Agriculture and precision farming*: UAVs monitored crop health using multispectral imaging, optimizing irrigation, pesticide use, and yield forecasting, contributing to sustainable agriculture practices.
- *Energy and infrastructure inspection*: Power lines, pipelines, and wind farms were routinely inspected using UAVs, reducing operational hazards and downtime compared to manned inspection methods.
- *Telecommunications and logistics*: Some drones were adapted for temporary communication relays in remote areas and experimental